

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 2**Date:** 1/15/24**Course:** Calculus Beta

Use implicit differentiation to find $\frac{dy}{dx}$.

$$4y^2 = \frac{3x - 2}{3x + 2}$$

To use implicit differentiation, first differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .

Use the rules of differentiation.

$$\frac{d}{dx}(4y^2) = \frac{d}{dx}\left(\frac{3x - 2}{3x + 2}\right)$$

To find $\frac{d}{dx}(4y^2)$, use implicit differentiation and the Power Rule for Positive Integers.

$$\frac{d}{dx}(4y^2) = 8y \frac{dy}{dx}$$

To find $\frac{d}{dx}\left(\frac{3x - 2}{3x + 2}\right)$, use the Derivative Quotient Rule.

$$\begin{aligned}\frac{d}{dx}\left(\frac{3x - 2}{3x + 2}\right) &= \frac{3(3x + 2) - 3(3x - 2)}{(3x + 2)^2} \\ &= \frac{12}{(3x + 2)^2}\end{aligned}$$

The terms with $\frac{dy}{dx}$ are already collected on one side of the equation.

$$8y \frac{dy}{dx} = \frac{12}{(3x + 2)^2}$$

Now, solve for $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{3}{2y(3x + 2)^2}$$

Thus, using implicit differentiation, $\frac{dy}{dx} = \frac{3}{2y(3x + 2)^2}$.